Shashank Shukla

Senior Platform Engineer

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SUMMARY

Platform, DevOps and SRE engineer with 5+ years designing, building and operating production infrastructure on AWS, GCP and Kubernetes. Architected a bring-your-own-cloud (BYOC) platform that stands up a complete customer environment, from cloud infrastructure and Kubernetes to 40–45 microservices, in 45 minutes. Runs 12 EKS clusters with 270+ microservices and leads a 4-engineer DevOps team, with depth in reliability, observability, cost optimization and security.

EXPERIENCE

Nielsen — Bangalore, India

Senior DevOps Engineer — Jun 2025 – Present

- Architected a bring-your-own-cloud (BYOC) delivery platform that builds a full production environment in a customer's AWS or GCP account from one config file: Pulumi infrastructure, Argo CD and Karpenter, then 40–45 microservices. Cut setup and upgrades from 6–7 hours to 45 minutes; live for 4 customers.
- Built AI-driven incident response on Amazon Bedrock: Prometheus alerts from 12 EKS clusters are triaged by models grounded in a Bedrock Knowledge Base of runbooks; common failures such as OOM kills are remediated automatically, and the rest reach on-call through incident.io with a diagnosis attached.
- Self-hosted two open-weight small language models on EKS, run as a fleet through a custom Kubernetes CRD, at ~4x lower inference cost than managed APIs.
- Operate 12 EKS clusters running 270+ microservices: zero-downtime upgrades from v1.32 to v1.34, NodeLocalDNS, Node Problem Detector, registry proxy cache and blue-green deployments; node memory swap cut OOM kills ~70%.
- Reduced AWS spend by ~$1,800/month by consolidating 20+ ALBs behind NGINX Ingress for 50+ services, and compute cost by ~30% through Graviton migration, multi-arch builds and Karpenter right-sizing.
- Built the real-time Ads platform infrastructure from scratch to production, designed for sub-10 ms responses on AWS Local Zones and Global Accelerator after ScyllaDB and Aerospike evaluations.
- Built the observability and on-call stack: Prometheus, Alertmanager, Loki and Grafana, with incident.io for alert routing, incident management and on-call rotations.
- Hardened cluster security with Kyverno policies (resource limits, no privileged containers, health and autoscaling checks) and Vault dynamic secrets via Kubernetes and AWS STS auth; led CrowdStrike remediation with patched-AMI rollouts across 15+ AWS accounts.
- Lead a 4-engineer DevOps team supporting 5–6 application teams; own GitLab CI, JFrog Artifactory with Vault integration, and the Docusaurus developer portal.

Conga — Bangalore, India

Senior DevOps Engineer — Apr 2024 – Jun 2025

- Led a 6-member DevOps/SRE team, setting deployment standards, runbooks and SRE practices, and ran CI/CD knowledge-sharing sessions across engineering teams.
- Cut deployment time from 25 minutes to under 8 across 200+ microservices, and brought per-release downtime under 30 seconds with blue-green deployments.
- Tripled deployment frequency and cut pipeline duplication 65% across 8+ teams with reusable Jenkins shared libraries and Spinnaker pipelines.
- Reduced release failures 70% and manual rollback incidents 85% by adding automated validation and auto-rollback to pipelines, and fixing session-persistence and schema-migration failures during traffic shifts.
- Cut manual infrastructure work 80% by automating environment provisioning with Terraform and Rancher Fleet, giving teams on-demand test environments.
- Made every deployment pass blocking quality gates: Checkmarx security scans, compliance checks and Slack approvals built into the pipeline.

- Designed event-driven autoscaling with KEDA for NGINX on Prometheus request metrics and for AWS SQS consumers on queue depth, replacing fixed replica counts.
- Built a fault-tolerant CRM data-migration orchestrator in Python and Celery on Kafka and MongoDB, with DAG dependencies, retries and a live dashboard; ran 500+ concurrent workflows with zero data loss.
- Built a SaaS API platform in FastAPI with 60+ endpoints, shipped through CI/CD with canary rollouts.

DevOps Engineer Apr 2023 – Mar 2024

- Delivered disaster recovery for 200+ microservices on Kubernetes with RTO under 10 minutes and RPO under 5 minutes through continuous replication and automated failover, with zero data loss across drills and real failovers.
- Cut release cycles from 2 weeks to 3 hours by automating highly available infrastructure with Terraform and adding artifact promotion, dynamic environments and rollback to pipelines.
- Cut MTTR by 50% by rolling out the observability stack (Prometheus, Grafana) and centralizing secrets in Doppler.
- Cut deployment time by 60% by standardizing CI/CD for React, Angular, Java and Python microservices.
- Cut AWS compute spend ~25% through Karpenter node autoscaling, cost audits and EC2 right-sizing workshops with service teams.
- Containerized legacy Java and Python services and migrated them to EKS with autoscaling and health probes.
- Hardened Terraform operations with S3-backed remote state, locking and versioning, and wrote Groovy and shell tooling for build orchestration, pipeline linting and secret rotation.

Associate DevOps Engineer Jul 2021 – Mar 2023

- Cut build times from 20 minutes to under 8 by parallelizing and caching Jenkins pipelines.
- Cut new-service setup from 4 to 2.4 hours with templatized Helm charts for stateless and stateful services, backed by automated chart validation and best-practice documentation.
- Built foundational Terraform modules for VPCs, IAM and EC2-based services across environments, supporting the move to cloud-native infrastructure.
- Enabled zero-downtime upgrades with load-balanced routing through CloudFront, Cloudflare and NGINX.
- Improved incident detection with centralized monitoring and logging on Grafana, Rancher and CloudWatch.

TECHNICAL SKILLS

- **Cloud:** AWS (EKS, EC2, ALB, RDS, DynamoDB, S3, SQS, IAM, STS, Bedrock, Local Zones), GCP
- **Kubernetes:** EKS, Karpenter, KEDA, Helm, Argo CD, NGINX Ingress, Kyverno, CRDs and Operators, Docker
- **IaC and CI/CD:** Pulumi, Terraform, GitLab CI, Jenkins, Spinnaker, Rancher Fleet, JFrog Artifactory, Blue-Green, Canary
- **Observability and SRE:** Prometheus, Alertmanager, Grafana, Loki, CloudWatch, incident.io, On-call, Disaster Recovery
- **AI and LLMOps:** Amazon Bedrock, Bedrock Knowledge Bases, Self-hosted SLMs on EKS, Runbook Automation
- **Security:** HashiCorp Vault, Kyverno, Checkmarx, Doppler, AWS STS, CrowdStrike
- **Networking:** NGINX, CloudFront, Cloudflare, AWS Global Accelerator, Local Zones
- **Languages and Data:** Python (FastAPI, Celery), Go, Bash, Groovy; Kafka, Redis, MongoDB, ScyllaDB, Aerospike

EDUCATION

B.Tech, Computer Science and Engineering — M.S. Ramaiah University of Applied Sciences Jul 2017 – Jun 2021
CGPA 9.0

CERTIFICATIONS

- Certified Kubernetes Administrator (CKA)
- HashiCorp Certified: Terraform Associate